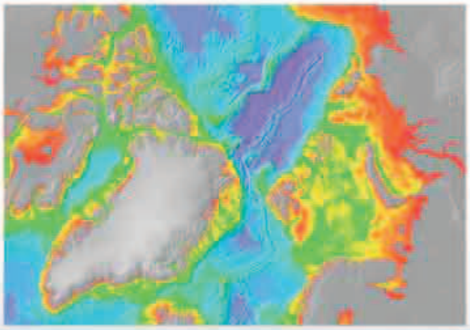
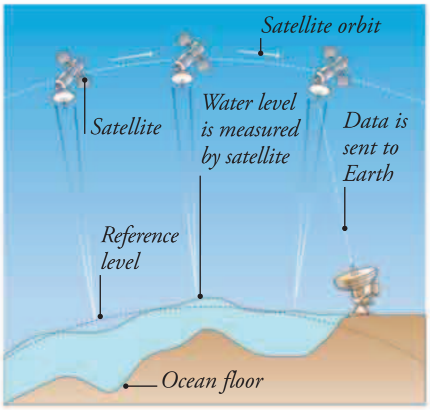
SONAR SURVEYS

The early research ships spent a lot of time measuring ocean depths using huge lengths of weighted cable. Today, however, research vessels use sonar technology, which produces a detailed image of the sea floor. Large areas of the ocean have been surveyed in this way, including this region around North Pole, color-coded for depth; the landmass is shown as gray.



EYES IN THE SKY



Orbiting satellites give us vital information about oceanic weather systems such as hurricanes. They can also map ocean currents, ice cover, water temperature, and plankton growth. Amazingly, measurements of the ocean surface compared to a reference level show that the water is not flat, but piles up above raised ocean floor features. Detected by satellites, these surface measurements can be used to create graphic, detailed images of the ocean floor.

WOW!

The maximum drilling depth of the *Chikyu* is greater than the height of Mount Everest, the world's highest mountain.